Laboratory 3:
Signals and Systems
CT/DT Systems and Basic System Properties
System Properties Explorer

Contents

Graded Assignment and Authenticity Requirement

Deliverable (1/13): This is a graded assignment. Include your name and student ID in all required screenshots and in as many additional screenshots as possible. Work missing clear student identification may lose credit.

Deliverable (2/13): All MATLAB verification must use your own captured input/output data from hardware. Synthetic data generated entirely in MATLAB is not accepted for property verification credit.

Checkpoint (1/8): Before submission, verify that all screenshots are legible and include axis labels, units, and measurement readouts where applicable.

Note on Prompts

Throughout this lab manual, four prompt types indicate required work.

- **Pre-lab Deliverable (N):** Work to complete before entering lab.
- **Checkpoint (N):** Evidence to show your TA during lab.
- **Discussion (N):** Analysis questions to answer in your submission.
- **Deliverable (N):** Required artifacts for grading.

1 Abstract

This laboratory builds a direct bridge between abstract system-property definitions and measurable hardware behavior. You will implement three physical systems driven by an Arduino source: a first-order RC filter, a diode clipper, and a comparator threshold circuit. For each system, you will evaluate linearity, time invariance, causality, and bounded-input bounded-output stability using both oscilloscope evidence and MATLAB-based numerical tests from captured data.

The instructional objective is not merely classification, but explanation: when a property fails, you must identify the mechanism (memory, thresholding, saturation, finite slew rate, or switching nonlinearity) responsible for the observed behavior.

2 Learning Objectives

By the end of this lab, you should be able to:

1. Map formal CT/DT property definitions to real circuit behavior under physical non-idealities.
2. Design and execute reproducible experiments to test linearity and time invariance using measured input/output pairs.
3. Distinguish theoretical causality/stability claims from practical measurement artifacts.
4. Quantify property-test agreement using error metrics (e.g., NRMSE) computed from acquired data.
5. Defend a system-property truth table with both mathematical argument and hardware evidence.

3 Theoretical Background

For a system operator $\mathcal{T}\{\cdot\}$ with input $x(t)$ and output $y(t)$:

- **Linearity:**
$$\mathcal{T}\{ax_1(t) + bx_2(t)\} = a\,\mathcal{T}\{x_1(t)\} + b\,\mathcal{T}\{x_2(t)\}$$
for all scalars a, b and admissible inputs x_1, x_2.
- **Time Invariance:**
$$x_\Delta(t) = x(t-\Delta) \Rightarrow \mathcal{T}\{x_\Delta(t)\} = y(t-\Delta)$$
where $y(t) = \mathcal{T}\{x(t)\}$.
- **Causality:** $y(t_0)$ depends only on input values at times $\tau \le t_0$.
- **BIBO Stability:**
$$|x(t)| \le M_x < \infty \Rightarrow |y(t)| \le M_y < \infty$$

For sampled data $x[n], y[n]$, use discrete analogs. To quantify test agreement, use normalized RMS error:

$$\text{NRMSE}(\%) = 100 \times \frac{\|y_{test}[n] - y_{ref}[n]\|_2}{\|y_{ref}[n]\|_2}$$

Small NRMSE supports property agreement over the tested operating region; it does not prove global validity.

4 Pre-Laboratory Assignment

Pre-lab Deliverable (1/2): Complete all pre-lab work before arriving. Bring handwritten derivations and predicted truth table.

4.1 Part A: Property Predictions

For each system below, predict whether it is linear, time-invariant, causal, and BIBO stable. Justify each claim in 2–4 technical sentences.

1. RC low-pass filter (first-order passive network)
2. Diode clipper (single-supply threshold clipping)
3. Comparator (open-loop threshold detector)

4.2 Part B: Test Signal Design

Pre-lab Deliverable (2/2): Create your own unique test set using your assigned random seed.Design and tabulate:

- Two base signals $x_1(t), x_2(t)$ (distinct shapes/frequencies)
- Scalars a, b for superposition test
- Time shift Δ (ms)
- Bounded amplitude limits for stability tests

4.3 Part C: MATLAB Verification Plan

Write pseudocode for:

1. importing captured CSV/serial data,
2. aligning time vectors and resampling if needed,
3. computing superposition residual and NRMSE,
4. computing time-invariance residual and NRMSE.

Deliverable (3/13): Submit scanned handwritten pre-lab pages with your name and student ID visible on the first page.

5 Equipment and Setup

Table 1. Required Hardware and Software

Category	Required Items
Hardware Source	Arduino (Uno/Mega or approved equivalent), USB cable, breadboard, jumper wires
Circuit Components	Resistors, capacitors, signal diode (e.g., 1N4148), op-amp/comparator IC, potentiometer (threshold set)
Measurement	Oscilloscope or ADALM2000, multimeter, probes/ground clips
Software	Arduino IDE, MATLAB (Signal Processing and basic plotting capability)
Documentation	Lab notebook pages for handwritten property truth tables

5.1 Assigned Parameters

Table 2. TA-Assigned Experiment Parameters

Symbol	Description	Assigned Value
f_1	Frequency for x_1	______________
f_2	Frequency for x_2	______________
a, b	Superposition coefficients	______________
Δ	Time shift for TI test	______________
A_{max}	Maximum allowed input amplitude	______________
Seed	Unique randomization seed	______________

Deliverable (4/13): Capture one physical setup photo that includes all three circuits (or three separate photos), each with your student ID visible in-frame.

6 Laboratory Procedure

6.1 Phase 1: Circuit Construction and Baseline Validation

6.1.1 Step 1.1 RC Filter Assembly

Checkpoint (2/8): Build and verify a first-order RC low-pass circuit driven by Arduino output.

Procedure:

1. Construct RC filter and verify wiring against your schematic.
2. Apply sinusoidal input and capture $x(t)$ and $y(t)$ on two channels.
3. Record gain and phase shift at your assigned frequency.

Deliverable (5/13): Submit dual-channel scope screenshot showing input/output waveforms with measured amplitude ratio and time delay annotations.

Step 1.1 Evidence: RC filter scope capture with ID visible

6.1.2 Step 1.2 Diode Clipper Assembly

Checkpoint (3/8): Build the clipper and demonstrate amplitude limiting behavior.

Procedure:

1. Build a diode clipping circuit for positive or negative threshold clipping.
2. Sweep input amplitude while observing output flattening/clipping onset.
3. Record approximate clipping threshold from measurement.

Deliverable (6/13): Submit scope screenshot with clipped waveform and labeled threshold estimate.

Step 1.2 Evidence: Diode clipper scope capture with ID visible

6.1.3 Step 1.3 Comparator Circuit Assembly

Checkpoint (4/8): Build comparator with adjustable threshold and verify binary switching behavior.

Procedure:

1. Set threshold using potentiometer or reference divider.
2. Apply analog input and observe output transitions.
3. Measure switching threshold and discuss hysteresis (if present).

Deliverable (7/13): Submit scope screenshot showing analog input and digital-like comparator output with threshold annotation.

Step 1.3 Evidence: Comparator scope capture with ID visible

6.2 Phase 2: Property Testing with Hardware Data

6.2.1 Step 2.1 Linearity Test (Superposition)

Checkpoint (5/8): For each circuit, acquire data to compare $\mathcal{T}\{ax_1 + bx_2\}$ versus $a\mathcal{T}\{x_1\} + b\mathcal{T}\{x_2\}$.

Required acquisition set per circuit:

1. Input/output for x_1
2. Input/output for x_2
3. Input/output for $ax_1 + bx_2$

Deliverable (8/13): Provide MATLAB plot overlay and NRMSE table quantifying superposition agreement for all three circuits.

Step 2.1 Evidence: MATLAB linearity overlays and NRMSE summary

Discussion (1/7): Explain why linearity is expected to hold (or fail) for each circuit. Distinguish intrinsic circuit behavior from measurement noise.

6.2.2 Step 2.2 Time-Invariance Test

Checkpoint (6/8): For each circuit, compare response to shifted input with shifted response to original input.

Required acquisition set per circuit:

1. Input/output for $x(t)$
2. Input/output for $x(t-\Delta)$

Deliverable (9/13): Provide MATLAB overlay of $\mathcal{T}\{x(t-\Delta)\}$ and $y(t-\Delta)$ and report NRMSE for each circuit.

Step 2.2 Evidence: MATLAB TI overlays and NRMSE summary

Discussion (2/7): If a mismatch appears, determine whether it is from non-TI behavior, trigger alignment error, finite sampling effects, or threshold drift.

6.2.3 Step 2.3 Causality and Stability Checks

Checkpoint (7/8): Execute bounded test inputs and confirm no anticipatory output behavior.

Procedure:

1. Apply bounded step/chirp/sinusoid inputs within $\pm A_{max}$.
2. Verify output remains bounded and note any saturation limits.
3. Use a delayed input onset and inspect output before onset to assess causality.

Deliverable (10/13): Submit scope evidence for causality check and bounded-output behavior for each circuit.

Step 2.3 Evidence: Scope captures for causality and BIBO checks

Discussion (3/7): State practical limits of your conclusions: tested amplitude range, bandwidth, and operating region.

6.3 Phase 3: MATLAB Verification from Captured Data

Checkpoint (8/8): Run your verification scripts using only acquired hardware data files.

6.3.1 Step 3.1 Data Import and Alignment

Deliverable (11/13): Submit MATLAB code snippets that import your captured datasets and align time bases for fair comparison.

Step 3.1 Evidence: MATLAB import/alignment code screenshot

6.3.2 Step 3.2 Numerical Property Metrics

Deliverable (12/13): Compute and report per-circuit NRMSE for linearity and time-invariance tests.

Table 3. Numerical Verification Summary

Circuit	Linearity NRMSE (%)	TI NRMSE (%)
RC filter	________________	________________
Diode clipper	________________	________________
Comparator	________________	________________

6.3.3 Step 3.3 Property Truth Table

Deliverable (13/13): Complete a handwritten truth table (Linear/TI/Causal/Stable) for each circuit and upload a scan/photo with your ID visible.

Table 4. System Property Truth Table (Final Conclusions)

System	Linear	TI	Causal	BIBO Stable	One-Sentence Justification
RC filter					
Diode clipper					
Comparator					

Step 3.3 Evidence: Handwritten truth table image with student ID

7 Discussion and Interpretation

Discussion (4/7): For each circuit, explain the dominant mechanism that determines property outcomes (memory, thresholding, clipping, or switching).

Discussion (5/7): Why can a system be causal and stable but still non-linear? Provide one example from your measured circuits.

Discussion (6/7): Describe one case where numerical verification might falsely suggest a property due to poor synchronization or insufficient input diversity.

Discussion (7/7): Connect your findings to model validity: when is an LTI approximation useful, and when does it fail for real hardware?

8 Final Submission Checklist

- ☐ Physical circuit photo(s) with student ID visible.
- ☐ Unique test input definitions (with assigned seed values).
- ☐ Scope evidence for RC, clipper, comparator baseline behavior.
- ☐ MATLAB linearity verification plots and NRMSE values.
- ☐ MATLAB time-invariance verification plots and NRMSE values.
- ☐ Causality and stability evidence captures.
- ☐ Handwritten property truth table with justifications and scope references.
- ☐ MATLAB scripts used for verification with captured data file references.

9 Grading Rubric (100 points)

Table 5. Lab 03 Evaluation Rubric

Category	Points	Criteria
Pre-lab quality	15	Correct predictions, clear derivations, and feasible test design.
Circuit implementation and evidence	20	Functional RC/clipper/comparator with clear annotated measurements.
Linearity verification	15	Proper superposition experiments with quantitative error analysis.
Time-invariance verification	15	Shift-based tests with correct alignment and numerical support.
Causality and stability evaluation	15	Convincing bounded-input and no-anticipation evidence.
Truth table and technical justification	10	Handwritten table accurate and defensible with data links.
Code quality and data authenticity	10	MATLAB uses captured data, reproducible workflow, readable scripts.